Oracle Belief Setup: Bounding the Value of Regime Information

Belief-Aware RL Project — Design Report v5

Belief-Aware RL Project

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1 Motivation

The sweep results in v1 and v4 establish a clear structural fact: which trading strategy performs best depends on the hidden composition of background agents, specifically the *ratio of momentum followers to value agents*.

- At `num_momentum_agents` = 0, `num_value_agents` = 102 (ratio $r = 0$): the VALUE/MR heuristic earns a median P&L of +328 cents while the MOMENTUM heuristic loses −1,967 cents (v4, Table 2).
- At `num_momentum_agents` = 12, `num_value_agents` = 102 (ratio $r \approx 0.12$, the default AAPL calibration): MOMENTUM earns +1,690 cents while VALUE loses −2,485 cents (v4, Table 2). The MOMENTUM heuristic also achieves a mean P&L of +66,172 cents and an 82% win rate over 50 episodes in the standalone baseline (v1, Table 3).
- At ratio $r \approx 0.49$ ($n_{\text{mom}} = 50$): both heuristics suffer catastrophic losses, indicating the regime has collapsed into a feedback-trading spiral that is harmful to all simple strategies.

This is the necessary and sufficient condition for a belief-aware agent to have an advantage: *different regimes reward different actions, and the agent observes neither the regime nor the agent counts directly.* The oracle setup formalizes this advantage as a testable upper bound.

2 Existing Data Supporting the Oracle Design

2.1 v4 PPO baseline (comparison target)

The belief-blind PPO agent trained on the default AAPL calibration ($n_{\text{mom}} = 12$, $n_{\text{val}} = 102$) across two seeds produces the metrics in Table 1. These numbers are the *pass/fail threshold* for all oracle experiments.

Quantity	**Seed 0**	**Seed 1**	**Avg.**
Mean final P&L (cents)	30,528	35,179	32,854
Std. final P&L (cents)	11,353	15,135	—
Sharpe ratio	2.689	2.324	2.507
Win rate	1.00	1.00	1.00
Mean max drawdown	0.00298	0.00341	0.00320
Timestep / episode length	60 s / 386 steps		
Eval episodes per seed	20		

Table 1: v4 belief-blind PPO baseline. Source: `results/ppo_baseline/seed_{0,1}/eval_metrics.json`. A third seed aborted during training; two-seed averages are used throughout.

2.2 Regime separation: targeted heuristic sweep

The targeted sweep (`targeted_sweep.json`; $n = 10$ episodes per cell, 300 s timestep / 78-step episodes) varies `num_momentum_agents` while holding $n_{\text{val}} = 102$ and all other parameters at default. Table 2 shows results for the two strategies directly relevant to the oracle design.

	VALUE strategy		MOMENTUM strategy	
n_{mom}	Median P&L	Win %	Median P&L	Win %
0 *(Val regime)*	+328	50	−1,967	30
12 *(Mom regime)*	−2,484	30	+1,690	60
50 *(excluded)*	−344,390	50	−1,561,646	0

Table 2: Heuristic strategy performance by momentum-agent count. All P&L values in cents; 300 s timestep, 10 episodes per cell. Timed-out episodes omitted for $n_{\text{mom}} = 50$. Source: `targeted_sweep.json`.

Key observations.

1. **Regime reversal is confirmed.** At $n_{\text{mom}} = 0$, VALUE outperforms MOMENTUM (+328 vs −1,967); at $n_{\text{mom}} = 12$, MOMENTUM outperforms VALUE (+1,690 vs −2,484). The sign of the advantage flips cleanly between the two oracle regimes.

2. **The extreme regime is excluded by design.** At $n_{\text{mom}} = 50$ both strategies suffer catastrophic losses; there is no useful action signal to condition on, so this setting is not used as an oracle target.

3. **Value-agent sweep is less clean.** Varying n_{val} while holding $n_{\text{mom}} = 12$ fixed shows a weaker crossover: at $n_{\text{val}} = 300$ both strategies lose (VALUE −2,740, MOMENTUM −3,318), consistent with over-anchoring collapsing the regime rather than creating a clean mean-reversion regime. The momentum-agent sweep is therefore the cleaner axis for defining oracle regimes.

2.3 E0 oracle heuristic estimate (sweep protocol)

Applying the oracle rule—play VALUE in Val episodes and MOMENTUM in Mom episodes—to the existing sweep data gives an *in-sweep* estimate of oracle heuristic performance:

Condition	n	**Mean P&L**	**Sharpe**	**Win %**
Oracle heuristic (Val: VALUE, Mom: MOMENTUM)	20	+1,182	0.15	55
Fixed MOMENTUM in both regimes	20	+647	0.08	45
v4 PPO baseline (60 s timestep)	40	+32,854	2.51	100

Table 3: Oracle heuristic estimate vs. v4 PPO baseline. **Warning:** heuristic rows use the 300 s / 78-step sweep protocol; the v4 baseline uses the 60 s / 386-step training protocol. Rows are *not* directly comparable — see Section 2.3.1.

2.3.1 Evaluation-protocol gap

The oracle heuristic mean P&L of +1,182 cents (Table 3) *cannot* be compared directly to the v4 PPO baseline mean of +32,854 cents. The two figures use different simulation protocols:

Setting	Sweep (heuristic)	PPO baseline
Timestep duration	300 s	60 s
Steps per episode	78	386
Episodes per cell	10	20
Train seeds	—	2

Table 4: Protocol differences between the targeted sweep and v4 PPO training.

Fewer steps per episode means fewer opportunities to earn (or lose) P&L. A rough linear scaling suggests the heuristic numbers should be multiplied by $386/78 \approx 5$ before comparing—but the relationship is nonlinear because large trending moves may be truncated at 78 steps. The proper E0 test (Section 7, step 1) must rerun the oracle heuristic under the *same 60 s / 386-step protocol* used by v4.

What the sweep does confirm. Protocol-corrected absolute P&L aside, the sweep establishes two protocol-independent facts: (a) the sign of the performance advantage flips between regimes for both strategies, and (b) the VALUE strategy under the oracle regime (+328 cents) already beats fixed MOMENTUM under the same regime (−1,967 cents) within the same protocol. These facts do not depend on the timestep, and they are sufficient to justify running E0 under the 60 s protocol.

3 Research Question

If the agent is given the true market composition as an auxiliary input, does it outperform the belief-blind PPO baseline?

If yes: regime information is worth estimating. Proceed to a learned belief module that infers the composition from public market history.
If no: the observable market signal is already sufficient, and a belief module adds no value.

We test this in two complementary ways, which we call the **two oracle variants**:

1. **Belief in observation.** Append the true regime belief tuple $b^* = (p_{\text{val}}, p_{\text{mom}})$ to the observation vector. The policy learns to use the ground-truth composition signal.
2. **Belief in reward.** Shape the step reward with a regime-alignment bonus that rewards the agent for taking actions consistent with the oracle regime. The base reward function is unchanged; only the training signal is augmented.

If *either* variant beats the v4 PPO baseline on both mean P&L and Sharpe ratio, the case for building a learned belief estimator is established.

4 Oracle Regime Configurations

4.1 Regime selection rationale

We restrict the oracle to **two regimes** derived directly from the existing sweep data, avoiding any new simulation runs unless the two-regime result is ambiguous.

Regime	Label	n_{mom}	n_{val}	$r = n_{\text{mom}}/n_{\text{val}}$	Favored heuristic
Value-dominated	VAL	0	102	0.000	MR / VALUE
Momentum-default	MOM	12	102	0.118	MOMENTUM

Table 5: Two oracle regime configurations. All other background parameters remain at the `rmsc04` AAPL calibration defaults.

The extreme-momentum setting ($n_{\text{mom}} = 50$, $r \approx 0.49$) is deliberately excluded: both heuristics lose heavily (median Momentum P&L = −1,561,646 cents, v4 Table 2), meaning the regime provides no useful action signal. An oracle over an incoherent regime is not a meaningful upper bound.

4.2 Oracle belief tuple

In each episode the environment is instantiated under one of the two regimes with equal probability: $P(\text{Val}) = P(\text{Mom}) = 0.5$. The oracle belief tuple b^* is a probability vector over these two classes:

$$b^* = \begin{cases} (1,\, 0) & \text{if episode is drawn from regime VAL,} \\ (0,\, 1) & \text{if episode is drawn from regime MOM.} \end{cases}$$

Under a hard oracle this is one-hot; under a soft oracle (used in the reward-shaping variant) it could be a temperature-smoothed distribution. For all v5 experiments we use the hard one-hot encoding. A soft version is deferred to the inferred-belief stage, where the estimator will produce calibrated probabilities rather than deterministic labels.

The tuple has dimension 2 and is appended to the existing 7-dimensional observation vector produced by `markets-daily_investor-v0`, producing an augmented observation of dimension 9.

5 Two Oracle Variants

5.1 Variant 1: belief in observation

Setup. The observation fed to the PPO policy at every step is:

$$\tilde{o}_t = [o_t \parallel b^*], \qquad o_t \in \mathbb{R}^7,\ b^* \in \{0,1\}^2,\ \tilde{o}_t \in \mathbb{R}^9.$$

The reward function is unchanged from the v4 PPO baseline (dense mark-to-market increment, scaled by the environment). The network architecture is the same as v4 (MLP policy).

What this tests. Whether the policy network can learn to condition its buy/hold/sell decisions on the regime label. An agent that correctly uses b^* should approach the performance of an oracle heuristic that always plays MOMENTUM in MOM episodes and MR in VAL episodes.

Expected outcome. Mean P&L and Sharpe should be strictly higher than the v4 baseline trained on the default AAPL calibration, because: (a) the VAL episodes now contribute positive MR P&L instead of the negative returns seen by a fixed momentum policy, and (b) the joint training distribution is enriched with the value-dominated regime, giving the policy experience at both ends of the ratio spectrum.

5.2 Variant 2: belief in reward

Setup. The observation is unchanged from the v4 baseline ($o_t \in \mathbb{R}^7$; the agent does *not* see b^*). The training reward is augmented with a regime-alignment term:

$$\tilde{r}_t = r_t + \alpha \cdot \text{align}(a_t,\, b^*),$$

where $\text{align}(a, b^*)$ is:

Action a_t	$\text{align}(a_t, \textsc{Val})$	$\text{align}(a_t, \textsc{Mom})$
Buy	-1	$+1$
Hold	0	0
Sell	$+1$	-1

Table 6: Regime-alignment bonus. The alignment encoding assumes: in VAL episodes the agent should fade moves (sell on recent buys, buy on recent sells); in MOM episodes it should follow pressure (buy on upward imbalance). Note: the table gives a simplified per-action version; a more accurate alignment could condition on the observation as well.

Calibrating α. The v4 baseline reports mean episode rewards of 7.8–9.0 over 386 steps, giving a typical *per-step* reward magnitude of $r_t \approx 0.018$–0.023. With $\text{align} \in \{-1, 0, +1\}$, the shaping bonus is $\alpha \cdot \text{align} \in [-\alpha, +\alpha]$. To avoid the bonus dominating the P&L signal:

$$\alpha \ll |r_t| \implies \alpha \ll 0.02 \implies \alpha \lesssim 0.002.$$

We therefore set $\alpha = 0.001$ as the default shaping coefficient. A search over $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}\}$ should be run once the training infrastructure is in place; $\alpha = 10^{-2}$ is the practical upper bound beyond which the shaping term exceeds r_t in magnitude and begins to corrupt the P&L credit signal.

What this tests. Whether the correct regime signal can be injected *through credit assignment* without the agent ever seeing the label at inference time. If the shaped policy converges to a better policy, it suggests the MDP reward signal alone is insufficient to distinguish regimes, and an auxiliary signal is needed for training even if not for deployment.

Evaluation. At evaluation time the alignment bonus is removed; the agent is scored solely on episode P&L and Sharpe under the standard reward. This makes the comparison to the v4 baseline fair.

6 Experimental Protocol

6.1 Training

All oracle variants train for the same budget as v4: 200,000 environment timesteps, 16 parallel runners, train batch size 2,000, five PPO epochs per batch, minibatch size 128, learning rate 10^{-4}, entropy coefficient 0.01.

Each variant is trained over **three seeds** (seeds 0, 1, 2); the failed third seed from v4 should be replicated here to achieve three-seed parity.

During training, each episode is drawn from either the VAL or MOM regime with equal probability, independently for each of the 16 parallel runners per batch.

6.2 Evaluation

Held-out evaluation uses **twenty episodes per regime per seed** (40 episodes total per seed: 20 VAL + 20 MOM), drawn from evaluation seeds 100–119 within each regime. This separates the regime-specific performance from the aggregate.

Condition	Obs dim	Reward
v4 PPO baseline (belief-blind)	7	MtM increment
Oracle variant 1 (belief in obs)	9	MtM increment
Oracle variant 2 (belief in reward)	7	MtM increment + $\alpha \cdot$ align (train only)

Table 7: Comparison conditions. All other hyperparameters are identical.

6.3 Pass/fail gate

The oracle experiment is declared successful—justifying the belief estimator stage—if *at least one variant* satisfies:

1. Mean P&L (aggregate across both regimes) > v4 mean P&L.
2. Sharpe ratio > v4 Sharpe ratio.
3. Win rate $\geq$ v4 win rate.

All three conditions must hold simultaneously. A variant that improves P&L but degrades Sharpe may indicate reward hacking rather than better regime use.

7 Testing Strategy

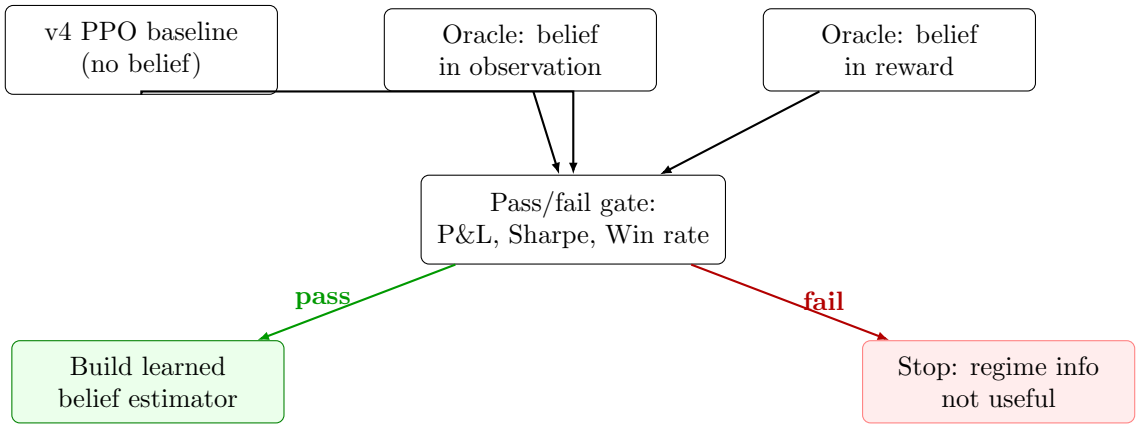

Figure 1: Decision tree for the oracle experiment. Passing at least one variant unlocks the belief-estimator stage.

7.1 Do we need to test it?

Yes, for one specific reason: the oracle is the *necessary* upper bound check. Without it, the project cannot distinguish between two failure modes:

1. The learned belief module is insufficiently accurate.
2. Regime information does not improve the policy at all (the observation signal already contains enough information for a fixed policy to perform well).

If the oracle fails the gate, failure mode (2) is the explanation and there is no point training a belief module. If the oracle passes, failure mode (1) is the relevant direction and the gap between the oracle agent and the learned-belief agent is the target to close.

7.2 Step-by-step test plan

1. **Heuristic sanity check (no training, ∼15 min).** Implement an oracle heuristic: play MOMENTUM if $b^* = (0, 1)$ and play MR if $b^* = (1, 0)$. Run 20 episodes per regime (40 total) under the *60 s / 386-step* protocol. The sweep already confirms regime reversal in sign (Section 2.3); this step establishes absolute P&L on the same footing as v4. **Status: pending** (sweep proxy exists; must rerun under 60 s protocol for direct v4 comparison).
2. **Train oracle variant 1 (belief in obs, 1.5 h per seed).** Augment the gym wrapper to append b^* to observations. Train with three seeds, evaluate as described in Section 6. Compare against v4 per-seed and aggregate.
3. **Train oracle variant 2 (belief in reward, 1.5 h per seed).** Implement a shaped reward wrapper. Train with three seeds. Evaluate with the shaping *removed* (standard reward only).
4. **Apply pass/fail gate.** If either variant passes, proceed to the inferred-belief stage (Section 9).
5. **Regime-stratified analysis.** Report metrics separately for VAL episodes and MOM episodes. A well-functioning oracle agent should improve in both regimes; a confused agent may improve in one while degrading in the other.

8 Oracle Experiment Results

Table 8 is the central results table for v5. Rows E0–E2 are filled in as experiments complete; E3 is the full belief-aware agent that follows once the gate passes.

Exp.	Agent	Mean P&L	Std P&L	Sharpe	Win %	Gate	Status
—	v4 PPO baseline (60 s)	32,854	—	2.51	100	*target*	complete
E0	Oracle heuristic (60 s)	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	pending
E1	PPO + belief in obs	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	pending
E2	PPO + belief in reward	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	pending
E3	PPO + inferred $\hat{b}$	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	blocked on E1/E2

Table 8: Oracle experiment results summary. Mean P&L and Std in cents; Sharpe and Win % are episode-level. Gate column: ✓ if experiment passes the pass/fail criteria in Section 6. "v4 PPO baseline" row uses the two-seed average from Table 1.

Regime-stratified breakdown (to be filled). Once E0 is complete, Table 9 should be populated.

	Val regime ($n_{\text{mom}} = 0$)			**Mom regime** ($n_{\text{mom}} = 12$)		
Agent	Mean P&L	Sharpe	Win %	Mean P&L	Sharpe	Win %
v4 PPO (belief-blind)	—	—	—	32,854	2.51	100
E0 oracle heuristic	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
E1 belief in obs	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
E2 belief in reward	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*

Table 9: Per-regime performance breakdown. The v4 PPO baseline was trained on the Mom regime only (default AAPL calibration); its Val-regime performance is unknown and should be measured as part of E0.

Note on v4 baseline regime coverage. The v4 PPO agent was trained and evaluated entirely on the default AAPL calibration ($n_{\text{mom}} = 12$, Mom regime). It has never seen a Val-regime episode ($n_{\text{mom}} = 0$). Whether it performs well or poorly in the Val regime is an open question; if it performs poorly (because it has learned a momentum-style policy), the oracle variants should show a clear improvement in that regime even before the Mom-regime comparison is made. This makes the Val-regime column in Table 9 particularly diagnostic.

9 From Oracle to Inferred Belief

Once the oracle gate passes, the b^* label is replaced by a learned estimator $\hat{b}_t = f_\phi(o_{t-k:t})$ that maps recent public market history to a soft belief vector over the two regimes.

Figure 2: Inference-time architecture once the belief estimator replaces the oracle. Δ^1 denotes the 2-simplex (probability vector over two regimes). The estimator f_ϕ can be a small LSTM or sliding window MLP trained to predict the regime label from public history.

Training the estimator. Generate labeled episodes: for each episode under regime $R \in \{\textsc{Val}, \textsc{Mom}\}$, the target is the corresponding one-hot b^*. Train f_ϕ to minimize cross-entropy between $\hat{b}_t$ and b^* using only observations $o_{t-k:t}$ (no privileged information). A k-step window of $k = 5$ to $k = 20$ steps is a reasonable starting point; the best window length should be selected by held-out classification accuracy.

Evaluation. Compare the inferred-belief PPO agent against: (a) the v4 belief-blind baseline and (b) the oracle agent. The gap between (b) and the inferred-belief agent quantifies how much accuracy the estimator sacrifices; the gap between (a) and the inferred-belief agent quantifies the practical gain from the belief module.

10 Summary of Proposed Experiments

Exp.	Agent type	Obs	New training?	Prerequisite
E0	Oracle heuristic (MR/Mom rule)	b^* visible	No	v1, v4 data
E1	PPO + belief in obs	$[o_t \| b^*]$	Yes	E0 passes
E2	PPO + belief in reward	o_t only	Yes	E0 passes
E3	PPO + inferred belief $\hat{b}_t$	$[o_t \| \hat{b}_t]$	Yes (+ estimator)	E1 or E2 passes gate

Table 10: Proposed experiment sequence. E0 is a fast sanity check; E1/E2 are the oracle variants; E3 is the full belief-aware RL agent.

The regime configurations (Table 5) are fully supported by existing v1 and v4 sweep data. No additional simulation runs are required unless E0 fails to show separation between the two regimes at the heuristic level, in which case an intermediate-ratio regime (e.g. $n_{\text{mom}} = 6$, $r \approx 0.06$) should be tested to find a cleaner boundary.